Ege Özgür

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Education

- **The University of Warwick** — Coventry, United Kingdom
  MSc Games Engineering — *Sep. 2025 – Sep. 2026*
- **Sabancı University** — Istanbul, Turkey
  BSc Computer Science and Engineering; GPA: 2.96 / 4.00 — *Sep. 2021 – Jun. 2025*

Experience

- **Farmazon** — Istanbul, Turkey
  Software Engineer Intern – Mandatory Summer Internship — *Jul. 2024 – Sep. 2024*
  - **Feature Design and Implementation**: Independently designed and built a GitHub Pull Request Checker tool from concept to delivery using .NET MVC, defining the data flow, interface layout, and filtering logic before implementation. Integrated the GitHub API and restructured JSON responses to produce a clean, readable display informed by how developers actually scan pull request data.
  - **Cross-Functional Collaboration**: Worked within an Agile team, documenting feature specifications, presenting iterations for feedback, and adjusting design decisions based on stakeholder input — championing the feature from inception to live delivery.
- **Makswin** — Istanbul, Turkey
  Software and Application Tester – Voluntary Internship — *Jul. 2023 – Aug. 2023*
  - **Test Planning and Quality Assurance**: Wrote and executed detailed test plans for software and mobile applications, identifying edge cases, documenting reproduction steps, and providing structured feedback to development teams. Developed a methodical, detail-oriented approach to quality validation across multiple feature areas.
  - **Cross-Team Collaboration**: Operated within a fast-paced corporate environment, coordinating with engineering and product teams to communicate findings clearly and support feature readiness ahead of release.

Projects

- **Skyfire Uprising – Level Designer, Game Engine Design and Development (May 2026)**: Working as Level Designer for Level 6 of "Skyfire Uprising", a singleplayer alien-invasion game developed in Unreal Engine 5 by a course-wide team divided into seven level groups. Designing a spaceport and research facility FPS level with focus on player flow, spatial readability, pacing, traversal routes, landmarks, and gameplay encounter placement. Used Autodesk Fusion and Unreal Engine 5 tools for blockout, layout iteration, modular environment planning, and in-engine level implementation. Collaborating with a six-person group under Scrum, with rotating group leadership across sessions.
- **Physically Based Renderer – Advanced Computer Graphics (May 2026)**: Extended a C++ path tracer with light tracing and instant radiosity rendering modes. Implemented multiple BSDFs (Diffuse, Mirror, Conductor, Glass, Oren-Nayar, Plastic, Layered), multiple importance sampling with the balance heuristic, and luminance-weighted environment map sampling. Built a binned SAH bounding volume hierarchy for acceleration, tile-based multithreading with persistent worker threads (achieving a 12.5x render-stage speedup on the Cornell box), and integrated Intel Open Image Denoise (OIDN) with albedo and normal auxiliary buffers. Compared performance and convergence behaviour of all three rendering modes across multiple test scenes. Project available on GitHub: github.com/ege-ozgur/RTBase-Renderer
- **Critical Review: Nanite vs. Traditional LOD Techniques (April 2026)**: Authored an academic review paper comparing Unreal Engine's Nanite virtualized geometry system with traditional Level of Detail techniques in real-time rendering. Analysed performance trade-offs, workflow complexity, adoption patterns, and technical limitations across discrete, continuous, view-dependent, and hierarchical LOD methods. Evaluated literature on quadric error metrics, progressive meshes, and Nanite's fallback mesh dependencies to argue for hybrid geometry-management pipelines in modern game development.
- **Tactical Five – Football Strategy Game (January 2026)**: Designed a football strategy game focused on decision-making rather than real-time control. Implemented card-based gameplay systems, possession mechanics, and probability-driven outcomes. Applied MDA framework and emergence theory to create strategic depth and improve player experience.

- **3D Game Development with DirectX 12 (December 2025)**: Developed a custom 3D game using C++ and DirectX 12. Implemented rendering pipeline setup, PSO management, constant buffers, shader reflection, instancing, animated and static meshes, camera system, input handling, and real-time performance considerations. Project available on GitHub: github.com/ege-ozgur/3D-Game-Directx12
- **3D Survivor Game (C++) (November 2025)**: Developed a 3D survivor-style game using C++. Implemented core gameplay mechanics including player movement, enemy AI behavior, collision detection, camera system, and real-time update loop. Project available on GitHub: github.com/ege-ozgur/3D-SurvivorGame
- **Central LA Application System (Sep. 2024 – Jun. 2025)**: Graduation project developed by a team of five for Sabancı University. Collaborated with faculty members to refine requirements and improve usability. Developed backend architecture using PHP and contributed to iterative feature development.
- **GitHub Pull Request Checker App (Jul. 2024 – Sep. 2024)**: Built a web tool for filtering and viewing pull requests using .NET MVC and HTML. Integrated GitHub API and modified JSON structure to create a clean and optimized display format.
- **Flight Management App (Feb. 2024 – Jun. 2024)**: Created domain model and sequence diagrams, designed PostgreSQL database schema, and implemented backend logic and APIs using Node.js.
- **FUTCARDS Mobile App (Sep. 2023 – Jan. 2024)**: Developed an app displaying FIFA-style football player cards. Built backend using Java and Spring, and implemented Android front-end using Android Studio.
- **Air Hockey Game on FPGA (Sep. 2023 – Jan. 2024)**: Designed ASM charts and implemented game logic using Verilog. Created test benches and deployed on an FPGA board using Vivado.
- **Natural Disasters Database Project (Feb. 2023 – Jun. 2023)**: Designed ER diagrams and built a MySQL database with tables, views, procedures, and triggers. Created visual analytics using Python and matplotlib.

Skills

- **Programming Languages**: C++, C#, JavaScript, Python, PHP, SQL
- **Frameworks & Tools**: Node.js, .NET MVC, Visual Studio, Visual Studio Code
- **Databases**: PostgreSQL, MySQL
- **Game Development**: C++ Game Architecture, Tile-based Rendering, DirectX12, Computer Graphics
- **Other**: Git/GitHub, Microsoft Office, Agile Development
- **Languages**: Turkish (Native), English (C1)

Interests

- **Football**: Playing, watching, and playing Football Manager and EA FC
- **Collecting**: Football jerseys, stickers, trading cards, and magnets
- **Other**: Video Games, Travelling